QUESTION BANK			
Academic Year	2025-26 (Odd Semester)	Scheme	Autonomous Scheme - 2024
Course Code	BCSOS303	Course Title	Operating System
Semester	III (3rd)	Department	Computer Science & Engineering/ Computer Science & Engineering(AI & ML)
Prepared By	1. 2. 3. 4.	Signatures	1. 2. 3. 4.
Verified By	1. 2.	Signatures	1. 2.
Approved By (HOD)		Signature	

Course Assessment Plan Matrix:

CO	Marks Distribution					Total Marks	Weightage (%)
	Module- 1	Module-2	Module-3	Module-4	Module-5		
CO1	20	5		5		30	30%
CO2		15	20		10	45	45%
CO3				15	10	25	25%
	20	**20**	**20**	**20**	**20**	**100**	**100%**

Choice of Answering in Modules	
Module No. 2	Module No. 4

Course Content Details:

MODULE-1	MODULE-2
1. What Operating Systems do 2. Operating System structure 3. Operating System operations 4. Operating System services, System Calls, System Programs 5. Design and Implementation	1. Process concept 2. Process scheduling 3. Operations on processes 4. Inter process communication 5. Overview of Threads 6. Multi-threaded models 7. Basic concepts of scheduling

6. Operating System structure– Simple, Layered, 7. Layered, Micro kernels and Modules **8.** Microkernels and Modules	8. Scheduling criteria and algorithms
MODULE-3 1. Process Synchronization Background 2. Critical section problem 3. Critical section problem 4. Peterson's solution 5. Semaphores 6. Semaphores 7. Classic problems of synchronization 8. Synchronization problems	**MODULE-4** 1. Background 2. Swapping, Contiguous memory allocation 3. Segmentation, paging 4. Structure of page table, Virtual memory 5. Background, Demand paging 6. Copy on write, Basic Page replacement 7. Page replacement algorithms: FIFO 8. Optimal and LRU.
MODULE-5 1. Background 2. System Model 3. Deadlock characterization 4. Methods for handling deadlocks 5. Deadlock prevention 6. Deadlock avoidance: Banker's algorithm 7. Deadlock detection 8. Deadlock detection	

Instructions to Students (To be Printed on the Question Paper):
1.
2.
3.
4.

MODULE 1					
Q. No.	**No.**	**Questions**	**Marks**	**BL-CO-PO**	**Topic**
1(a)	1	Explain the user's view and system's view of an operating system.	10	L2-CO1-PO1	OS role
	2	Explain the following with an example: i) Time sharing system ii) Job scheduling	10	L2-CO1-PO1	OS structure
	3	Explain the abstract view of the components of a computer system with a neat diagram.	10	L2-CO1-PO1	OS role

	4	Briefly explain multiprogramming system with a neat diagram.	10	L2-CO1-PO1	OS structure
	5	Explain the operating system services which provide functions that are helpful to the user	10	L2-CO1-PO1	OS services
	6	Explain dual mode and multimode operation of an operating system with a neat diagram.	10	L2-CO1-PO1	OS Operations
	7	Describe with an example, how timer operation is used to compute current timer of the system.	10	L2-CO1-PO1	OS Operations
	8	Explain User mode and Kernel Mode of an Operating system.	10	L2-CO1-PO1	OS Operations
	9	Briefly explain operating system services to ensure efficient operation of the system with a neat diagram.	10	L2-CO1-PO1	OS Services
	10	The issues of the resource utilization show up in different forms in different types of operating systems. List what resources must be managed carefully in the following settings: a) Mainframe or Minicomputer systems. b) Workstations connected to servers. c) Mobile Computers.	10	L2-CO1-PO1	Introduction to OS
1(b)	1	Explain the purpose of system calls and their types with suitable examples.	10	L2-CO1-PO1	System Calls
	2	Explain the role of system call interface and passing of parameters as a table	10	L2-CO1-PO1	System calls
	3	Explain the process of copying the content of one file to another sfile using different system calls.	10	L2-CO1-PO1	System Programs
	4	Explain layered approach of operating system with a neat diagram.	10	L2-CO1-PO1	Operating Services
	5	Explain the following with pictorial representations: i. MS-DOS layered structure ii. Traditional UNIX system structure	10	L2-CO1-PO1	Structure of Operating system
	6	Explain the advantages and disadvantages of the layered approach.	10	L2-CO1-PO1	Layered Approach
	7	Explain the architecture of a typical microkernel with neat sketch.	10	L2-CO1-PO1	Micro kernel Architecture
	8	Describe the role of loadable kernel modules, and illustrate the Solaris loadable modules with the help of a neat sketch.	10	L2-CO1-PO1	Modules
	9	Explain the following considerations while building an operating system: i.Design goals with mechanism-policy separation ii. Implementation methods	10	L2-CO1-PO1	Design and Implementation

<table>
<tr><td></td><td>10</td><td>Explain process control system call. explain MS-DOS execution at the time of system startup and during running a program with a diagram.</td><td>10</td><td>L2-CO1-PO1</td><td>API ,System calls and operating system</td></tr>
<tr><td colspan="6">MODULE 2 (Choice of Answering)</td></tr>
<tr><td rowspan="10">2(a)</td><td>1</td><td>Explain various states of a process with state transition diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Process Concept</td></tr>
<tr><td>2</td><td>Explain Process Control Block with a neat diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Process Concept</td></tr>
<tr><td>3</td><td>Explain with diagram how CPU switch from process to process.</td><td>5</td><td>L2-CO1-PO1</td><td>Process Concept</td></tr>
<tr><td>4</td><td>Describe Scheduling Queues with a neat Queuing-diagram of process scheduling.</td><td>5</td><td>L2-CO1-PO1</td><td>Process Scheduling</td></tr>
<tr><td>5</td><td>Explain Indirect communication with respect to message passing systems.</td><td>5</td><td>L2-CO1-PO1</td><td>Interprocess Communication</td></tr>
<tr><td>6</td><td>With a block diagram explain process creation using the fork() system call.</td><td>5</td><td>L2-CO1-PO1</td><td>Operations on Processes</td></tr>
<tr><td>7</td><td>Write a C program to create a separate process using fork() system call.</td><td>5</td><td>L2-CO1-PO1</td><td>Operations on Processes</td></tr>
<tr><td>8</td><td>Explain Single-threaded and multithreaded processes with diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Overview of Threads</td></tr>
<tr><td>9</td><td>Outline the benefits of Multithreaded programming.</td><td>5</td><td>L2-CO1-PO1</td><td>Overview of Threads</td></tr>
<tr><td>10</td><td>Explain any two multithreading models.</td><td>5</td><td>L2-CO1-PO1</td><td>Multithreading Models</td></tr>
<tr><td>2(b)</td><td>1</td><td>Consider the following set of processes with the length of CPU burst time given in milliseconds:
<table>
<tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr>
<tr><td>P1</td><td>0</td><td>5</td><td>3</td></tr>
<tr><td>P2</td><td>1</td><td>3</td><td>1</td></tr>
<tr><td>P3</td><td>2</td><td>8</td><td>2</td></tr>
<tr><td>P4</td><td>3</td><td>6</td><td>4</td></tr>
</table>
a) Draw Gantt charts for these processes using the following scheduling algorithms:
I. FCFS
II. SJF (Preemptive)
III. Non-Preemptive Priority (1 = highest priority)
IV. RR (Time quantum = 2 ms)</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
</table>

<table>
<tr><td></td><td></td><td>b) Calculate Average Waiting Time (AWT) and Average Turnaround Time (ATT) for each scheduling algorithm.
c) Identify which algorithm gives the minimum AWT.</td><td></td><td></td><td></td></tr>
<tr><td></td><td>2</td><td>Consider the following processes:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>1</td><td>4</td></tr><tr><td>P2</td><td>0</td><td>6</td></tr><tr><td>P3</td><td>3</td><td>8</td></tr><tr><td>P4</td><td>0</td><td>3</td></tr><tr><td>P5</td><td>2</td><td>5</td></tr></table>a) Apply FCFS, SJF (Preemptive) and RR with TQ= 2ms for the above table.
b) Calculate Average Waiting Time and Average Turnaround Time.
c) Compare the results.</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>3</td><td>Given the process table:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>4</td><td>6</td><td>3</td></tr><tr><td>P2</td><td>2</td><td>4</td><td>1</td></tr><tr><td>P3</td><td>0</td><td>5</td><td>2</td></tr><tr><td>P4</td><td>1</td><td>5</td><td>1</td></tr></table>a) Use Preemptive Priority Scheduling (lowest number = highest priority), SJF (preemptive), FCFS and draw the Gantt charts.
b) Calculate Average Waiting Time and Average Turnaround Time.
c) Identify which algorithm gives the minimum AWT.</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>4</td><td>Three processes arrive as follows:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>0</td><td>12</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>3</td><td>5</td></tr><tr><td>P4</td><td>2</td><td>8</td></tr><tr><td>P5</td><td>4</td><td>3</td></tr></table>a) Schedule the processes using Round Robin with TQ=4 ms, FCFS and Preemptive SJF. Also draw the Gantt chart for the same.
b) Calculate Average Waiting Time and Average Turnaround Time.
c) How would results change if TQ=6 ms for RR?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>5</td><td>Given the following CPU burst.
<table><tr><th>Process</th><th>Arrival Time</th><th>CPU Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>3</td><td>4</td></tr><tr><td>P2</td><td>1</td><td>1</td><td>2</td></tr><tr><td>P3</td><td>3</td><td>2</td><td>3</td></tr><tr><td>P4</td><td>4</td><td>5</td><td>1</td></tr></table>a) Draw Gantt charts and compute Average Waiting Time and Average Turnaround Time.</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
</table>

<table>
<tr><td></td><td></td><td>b) Compare FCFS, Preemptive SJF, Non-preemptive Priority and Round Robin scheduling (TQ=2ms).
c) Which is optimal for minimizing waiting time?</td><td></td><td></td><td></td></tr>
<tr><td></td><td>6</td><td>Five processes arrive as shown:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>8</td><td>2</td></tr><tr><td>P2</td><td>1</td><td>4</td><td>1</td></tr><tr><td>P3</td><td>2</td><td>9</td><td>3</td></tr><tr><td>P4</td><td>3</td><td>5</td><td>4</td></tr><tr><td>P5</td><td>4</td><td>2</td><td>5</td></tr></table>a) Draw Gantt charts and compute Average Waiting Time and Average Turnaround Time.
b) Compare FCFS, Preemptive SJF, and Non-preemptive Priority scheduling.
c) Which is optimal for minimizing waiting time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>7</td><td>Consider the following processes with their arrival time, burst time, and priority:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>9</td><td>2</td></tr><tr><td>P2</td><td>1</td><td>5</td><td>1</td></tr><tr><td>P3</td><td>2</td><td>7</td><td>3</td></tr><tr><td>P4</td><td>3</td><td>4</td><td>2</td></tr></table>a) Draw the Gantt charts for the above processes using FCFS, Preemptive SJF, Non-preemptive Priority, and Round Robin (TQ=3 ms).
b) Calculate Average Waiting Time and Average Turnaround Time for each algorithm. Which algorithm gives minimum Average Waiting Time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>8</td><td>Consider the following processes with their arrival time and burst time.
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>0</td><td>3</td></tr><tr><td>P2</td><td>1</td><td>6</td></tr><tr><td>P3</td><td>2</td><td>4</td></tr><tr><td>P4</td><td>3</td><td>5</td></tr><tr><td>P5</td><td>4</td><td>2</td></tr></table>a) Simulate execution using FCFS, Round Robin (TQ=2 ms), and Preemptive SJF. Draw Gantt charts for each case
b) Calculate Average Waiting Time and Average Turnaround Time. Which algorithm gives minimum Average Waiting Time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>9</td><td>Given 4 processes:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>10</td><td>1</td></tr><tr><td>P2</td><td>1</td><td>8</td><td>2</td></tr><tr><td>P3</td><td>2</td><td>6</td><td>4</td></tr><tr><td>P4</td><td>3</td><td>4</td><td>3</td></tr></table></td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
</table>

<table>
<tr><td></td><td></td><td>a). Draw the Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, Non-Preemptive SJF, preemptive priority (Lower number = higher priority), and RR (quantum = 5ms).
b). Compute the average turnaround time and average waiting time of processes for each of those scheduling algorithms.
c). Compare the results</td><td></td><td></td><td></td></tr>
<tr><td></td><td>10</td><td>Consider the following set of processes, with the length of the CPU burst given in milliseconds:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>12</td><td>3</td></tr><tr><td>P2</td><td>2</td><td>6</td><td>1</td></tr><tr><td>P3</td><td>4</td><td>8</td><td>4</td></tr><tr><td>P4</td><td>5</td><td>5</td><td>2</td></tr></table>a). Draw the Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, pre-emptive priority (Lower number = higher priority), and RR (quantum = 3).
b). Compute the average turnaround time and average waiting time of processes for each of those scheduling algorithms.
c). Which algorithm results in minimum average waiting time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td rowspan="9">3(a)</td><td>1</td><td>Explain different CPU Scheduling criteria’s.</td><td>5</td><td>L2-CO1-PO1</td><td>Scheduling Criteria</td></tr>
<tr><td>2</td><td>Define the following:
a) Shortest-Job-First Scheduling algorithm
b) Round robin scheduling algorithm
Priority scheduling algorithm</td><td>5</td><td>L2-CO1-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td>3</td><td>Define Thread. Explain the concept of thread with suitable example.</td><td>5</td><td>L2-CO1-PO1</td><td>Overview of Threads</td></tr>
<tr><td>4</td><td>Explain long-term and short-term schedulers.</td><td>5</td><td>L2-CO1-PO1</td><td>Basic Concepts of Scheduling</td></tr>
<tr><td>5</td><td>Describe the working of process termination.</td><td>5</td><td>L2-CO1-PO1</td><td>Operations on Processes</td></tr>
<tr><td>6</td><td>Define the concept of medium term scheduling with the help of queuing diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Process Scheduling</td></tr>
<tr><td>7</td><td>Explain multithreaded server architecture with a neat diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Overview of Threads</td></tr>
<tr><td>8</td><td>Explain the alternating sequence of CPU and I/O bursts with a neat diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Basic Concepts of Scheduling</td></tr>
<tr><td>9</td><td>Explain shared memory communication model with a neat diagram.</td><td>5</td><td>L2-CO1-PO1</td><td>Inter process Communication</td></tr>
</table>

<table>
<tr><td></td><td>10</td><td>Write a snippet for Producer process and Consumer process using shared memory.</td><td>5</td><td>L2-CO1-PO1</td><td>Inter process Communication</td></tr>
<tr><td rowspan="4">3(b)</td><td>1</td><td>Five processes p1, p2, p3, p4 and P5 have burst time 7, 9, 6, 7, 10 and process have arrived at 0, 5, 2, 7 and 4 milli seconds respectively. Draw Gantt chart to illustrate the execution of these processes using FCFS, RR scheduling (Quantum = 2ms) and preemptive SJF scheduling. Calculate average turnaround time and average waiting time for each of the scheduling algorithms. Also, which algorithm results in minimum average waiting time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td>2</td><td>Consider the following set of processes with the length of CPU burst time given in milliseconds.
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>10</td><td>4</td></tr><tr><td>P2</td><td>3</td><td>5</td><td>1</td></tr><tr><td>P3</td><td>3</td><td>6</td><td>6</td></tr><tr><td>P4</td><td>5</td><td>4</td><td>3</td></tr></table>
a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms i) FCFS ii) SJF iii) Non-Preemptive Priority (Consider 1 as High Priority) iv) RR (Time quantum=2)
b) Find the average waiting time and turnaround time for each scheduling algorithm.
c) Which algorithm results in minimum average waiting time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td>3</td><td>Given the process set:
<table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>7</td><td>1</td></tr><tr><td>P2</td><td>2</td><td>5</td><td>3</td></tr><tr><td>P3</td><td>4</td><td>4</td><td>2</td></tr><tr><td>P4</td><td>1</td><td>5</td><td>4</td></tr><tr><td>P5</td><td>3</td><td>3</td><td>2</td></tr></table>
a) Apply Round Robin with TQ=2 ms and TQ=5 ms. Draw both Gantt charts.
b) Compare AWT and ATT for both cases.
c) Which time quantum is better here?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td>4</td><td>Five processes P1, P2, P3, P4, and P5 have the following burst times and arrival times:
<table><tr><th>Process</th><th>Arrival Time(ms)</th><th>Burst Time(ms)</th></tr><tr><td>P1</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>3</td><td>8</td></tr><tr><td>P3</td><td>1</td><td>4</td></tr><tr><td>P4</td><td>5</td><td>6</td></tr><tr><td>P5</td><td>2</td><td>7</td></tr></table>
a) Draw the Gantt chart to illustrate the execution of these processes using: FCFS, Round Robin (Quantum = 3 ms), Preemptive Shortest Job First (SJF)</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
</table>

<table>
<tr><td></td><td></td><td>b) Also calculate Average Turnaround Time and Average Waiting Time for each algorithm.

c) Which algorithm results in minimum average waiting time?</td><td></td><td></td><td></td></tr>
<tr><td></td><td>5</td><td>Consider the following set of processes, with the length of the CPU burst time given in milliseconds

<table><tr><td>Process</td><td>Burst Time</td><td>Priority</td></tr><tr><td>P1</td><td>2</td><td>2</td></tr><tr><td>P2</td><td>1</td><td>1</td></tr><tr><td>P3</td><td>8</td><td>4</td></tr><tr><td>P4</td><td>4</td><td>2</td></tr><tr><td>P5</td><td>5</td><td>3</td></tr></table>
a). Draw Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-pre-emptive priority (a larger number implies a higher priority).

b). Compute the average turnaround time and average waiting time of processes for each of those scheduling algorithms.

c). Which algorithm results in minimum average waiting time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>6</td><td>Consider the following set of processes, with the length of the CPU burst given in milliseconds:
<table><tr><td>Process</td><td>Arrival Time</td><td>CPU Burst</td><td>Priority</td></tr><tr><td>P1</td><td>5</td><td>3</td><td>1</td></tr><tr><td>P2</td><td>3</td><td>4</td><td>2</td></tr><tr><td>P3</td><td>4</td><td>7</td><td>3</td></tr><tr><td>P4</td><td>1</td><td>5</td><td>4</td></tr><tr><td>P5</td><td>2</td><td>2</td><td>2</td></tr></table>
a). Draw Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, preemptive priority (a larger number implies a higher priority) and RR with TQ=3ms.

b). Compute the average turnaround time and average waiting time of processes for each of those scheduling algorithms.
c). Which algorithm results in minimum average waiting time?</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>7</td><td>a) Illustrate the working of multilevel feedback queue scheduling.
b) Illustrate the working of multilevel queue scheduling.</td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
<tr><td></td><td>8</td><td>Consider the following set of processes with the length of CPU burst time given in milliseconds.
<table><tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td><td>Priority</td></tr><tr><td>P1</td><td>0</td><td>10</td><td>4</td></tr><tr><td>P2</td><td>1</td><td>6</td><td>1</td></tr><tr><td>P3</td><td>4</td><td>4</td><td>3</td></tr></table></td><td>15</td><td>L3-CO2-PO1</td><td>Scheduling Algorithms</td></tr>
</table>

		P4 | 5 | 3 | 2 a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms i) FCFS ii) SJF iii) Non-Preemptive Priority (Consider 1 as High Priority) iv) RR (Time quantum=2) b) Find the average waiting time and turnaround time for each scheduling algorithm. c) Which algorithm results in minimum average waiting time?				
	9	Consider the following set of processes, with the length of the CPU burst given in milliseconds: Process | Arrival Time | Burst Time | Priority P1 | 0 | 6 | 4 P2 | 3 | 5 | 2 P3 | 3 | 3 | 6 P4 | 5 | 5 | 3 a). Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, preemptive priority (Lower number = higher priority), and RR (quantum = 1). b). Compute the average turnaround time and average waiting time for processes for each of those scheduling algorithms. c). Which algorithm results in minimum average waiting time?	15	L3-CO2-PO1	Scheduling Algorithms
	10	Four processes p1, p2, p3 and p4 having burst time of10, 5, 6, 7 have arrived at 0, 3, 4 and 5 milli seconds respectively. Priority of the processes are 2, 1, 2, and 4 (Lowest number = highest priority). Draw Gantt chart to illustrate the execution of these processes using FCFS, RR scheduling (Quantum = 2ms) and preemptive priority scheduling. Calculate average turnaround time and average waiting time for each of the scheduling algorithms. Also, which algorithm results in minimum average waiting time?	15	L3-CO2-PO1	Scheduling Algorithms
MODULE 3						
4(a)	1	Apply the process synchronization concept to a real-world problem along with the general structure of process synchronization.	10	L3-CO2-PO1	Process Synchronization
	2	Illustrate the way Semaphore overcomes the drawbacks of Peterson's solutions along with usage and implementation of semaphore with modified functions.	10	L3-CO2-PO1	Peterson's solutions
	3	Apply Peterson's Solution to the Critical section Problem.	10	L3-CO2-PO1	Peterson's solutions
	4	Apply Peterson's Solution along with the structure for process synchronization.	10	L3-CO2-PO1	Peterson's solutions
	5	Illustrate the critical section problem with an algorithm.s	10	L3-CO2-PO1	Critical section

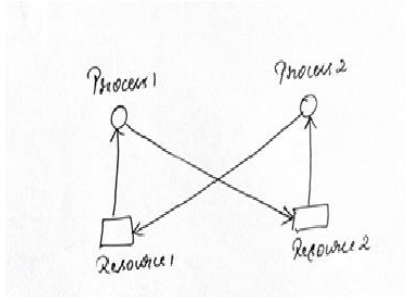
	6	Illustrate a scenario where race conditions might occur. How can such conditions be prevented?	10	L3-CO2-PO1	Process Synchronization
	7	Apply the process synchronization concept along with the general structure for Producer & consumer problem.	10	L3-CO2-PO1	Process Synchronization
	8	Illustrate the three requirements that must satisfy to solve the critical section problem by considering problem as an example.	10	L3-CO2-PO1	Critical section
	9	Consider a shared bank account balance accessed by multiple ATMs simultaneously. Explain how a race condition may occur and apply semaphore-based synchronization to ensure correct balance updates.	10	L3-CO2-PO1	Process Synchronization
	10	Apply the concepts of Critical section problem by taking example of online ticket booking system.	10	L3-CO2-PO1	Critical section
4(b)	1	Illustrate the application of semaphore to solve Bounded-Buffer problem along with the Snippets.	10	L3-CO2-PO1	Semaphores
	2	Illustrate the application of semaphore to solve Dining Philosophers problem.	10	L3-CO2-PO1	Semaphores
	3	Illustrate the application of semaphore to solve Readers-Writer's problem.	10	L3-CO2-PO1	Semaphores
	4	Apply the usage of Monitors to solve Readers-Writer's problem.	10	L3-CO2-PO1	Monitors
	5	Apply your understanding of semaphore operations to analyze how the incorrect ordering of wait() and signal() calls in a concurrent system can lead to deadlock. Illustrate your answer using a two-process scenario.	10	L3-CO2-PO1	Semaphores
	6	Illustrate structure of Monitors along with condition variables.	10	L3-CO2-PO1	Semaphores
	7	Implement a Monitor Using Semaphores.	10	L3-CO2-PO1	Semaphores
	8	Apply the concept of semaphores to a real-time system where a low-priority process holds a resource needed by a high-priority process. How would you detect and resolve priority inversion in this scenario?	10	L3-CO2-PO1	Semaphores
	9	Apply the ResourceAllocator monitor structure to demonstrate how a single shared resource can be safely allocated among multiple processes. Explain how conditional wait with priority values improves scheduling.	10	L3-CO2-PO1	Monitors
	10	Illustrate Semaphores with atomic operations and also types of semaphores.	10	L3-CO2-PO1	Semaphores

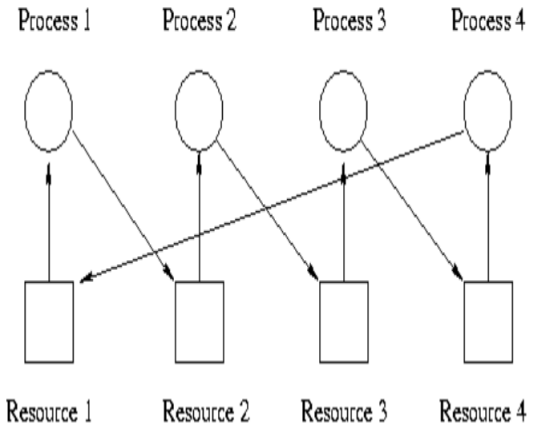
MODULE 4 (Choice of Answering)					
5(a)	1	Explain logical and physical address space.	5	L2-CO1-PO1	memory management
	2	Explain Swapping in memory management.	5	L2-CO1-PO1	memory management
	3	Explain Paging Concept.	5	L2-CO1-PO1	Paging
	4	Explain Contiguous Memory allocation.	5	L2-CO1-PO1	memory management
	5	Explain segmentation.	5	L2-CO1-PO1	segmentation
	6	Explain the concept of paging and segmentation.	5	L2-CO1-PO1	segmentation
	7	Explain Base register and limit register in the logical address space with an example.	5	L2-CO1-PO1	memory management
	8	Explain Address binding along with its different stages where address binding will be done.	5	L2-CO1-PO1	Memory Binding
	9	Explain dynamic linking and shared libraries.	5	L2-CO1-PO1	Dynamic Linking/ Loading
	10	Explain internal and external fragmentation	5	L2-CO1-PO1	fragmentation
5(b)	1	Assume if it takes 100 nanoseconds to access memory, then a mapped-memory access takes 100 nanoseconds when the page number is in the TLB, hit ratio is 80% and 200 nanoseconds if we fail to find the page number in the TLB. Calculate effective memory-access time. Analyze the improvement if hit ratio is 99%.	8	L4-CO3-PO2	TLB
	2	Consider a paging system with the page table stored in memory. a. If a memory reference takes 50 nanoseconds, how long does a paged memory reference take? b. If we add TLBs, and 75 percent of all page-table references are found in the TLBs, verify the change in effective memory reference time. (Assume that finding a page-table entry in the TLBs takes 2 nanoseconds, if the entry is present.)	8	L4-CO3-PO2	TLB
	3	Consider paging hardware with a TLB, assume that the entire page table and all the pages is in the physical memory, its takes 10ms to search the TLB and 80ms to	8	L4-CO3-PO2	TLB

		access the physical memory. If the TLB hit ratio is 0.6, what is the effective memory access time (in milli seconds).			
	4	Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. Verify, how many page faults would occur for the 3 and 4 frames using FIFO replacement algorithms, justify the **Belady's anomaly** concept**.**	8	L4-CO3-PO2	Paging
	5	Analyze the following reference string: 5,7,5,6,2,1,2,3,1,2,3,6,4,3,5,4 for a memory with four frames. Find the number of page faults for FIFO and optimal page replacement algorithms. Which is the most efficient among them?	8	L4-CO3-PO2	Paging
	6	Consider the following page reference string:7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0 , 1. Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms? a. LRU replacement. b. Optimal replacement.	8	L4-CO3-PO2	Paging
	7	A machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB, what is the approximate size of the page table?	8	L4-CO3-PO2	Paging
	8	Consider the following page reference string: 1,2,3,4,1,2,5,1,2,3,4,5. Calculate page faults by LRU and Optimal replacement with 4 frames.	8	L4-CO3-PO2	Paging
	9	A process requires 10 pages, each of size 1 KB. The system uses 12-bit logical addresses and a page size of 256 bytes. Analyze how the logical address is divided into page number and offset and explain the role of page table in this case.	8	L4-CO3-PO2	Paging
	10	Given a logical address space of 32 pages, each of size 1 KB, and a physical memory of 8 frames, analyze how paging achieves memory management and discuss the internal fragmentation issue in this scenario.	8	L4-CO3-PO2	Paging
5(c)	1	Compare and contrast Paging and Pure demand paging.	7	L4-CO3-PO2	Paging
	2	Compare and contrast the Shared memory and demand paging concepts in virtual memory.	7	L4-CO3-PO2	Paging
	3	Examine whether Effective Access Time in demand paging is directly proportional to page fault time or not.	7	L4-CO3-PO2	Paging
	4	Consider a process is trying to access a page that was not brought into memory. Analyse and illustrate the actions to be taken by Operating System to handle the scenario effectively, given that single level paging is used in the system.	7	L4-CO3-PO2	Paging

	5	Compare and contrast inverted page tables and hierarchical page tables.	7	L4-CO3-PO2	Paging
	6	The page fault service time be 08 milli secs in a computer with average memory access time being 200ns. If one page fault is generated for every 1000 memory accesses, what is the effective access time for the memory? Also, Verify the probability of page faults to have a performance degradation of less than 10%.	7	L4-CO3-PO2	Effective Memory access
	7	Compare and contrast hashed page table and inverted page table	7	L4-CO3-PO2	Paging
	8	The page fault service time be 04milli sec in a computer with average memory access time being 100nano sec. If one page fault is generated for every 500 memory accesses, what is the access time for the memory? Also verify the probability of the page fault to have a performance degradation of less than 20%.	7	L4-CO3-PO2	Effective Memory access
	9	Analyze how copy-on-write (COW) paging helps in efficient process creation, especially during the **fork()** system call.	7	L4-CO3-PO2	copy-on-write
	10	Consider the following segment table: Segment Base Length 0 219 600 1 2300 14 2 90 100 3 1327 580 4 1952 96 What are the physical addresses for the following logical addresses? a. 0,430 b. 1,10 c. 2,500 d. 3,400 e. 4,112	7	L4-CO3-PO2	Memory Address
OR					
6(a)	1	Explain Copy on write	5	L2-CO1-PO1	Paging
	2	Explain Memory Access time, along with Hit ratio and Miss ratio.	5	L2-CO1-PO1	TLB
	3	Explain Virtual memory and its advantages.	5	L2-CO1-PO1	memory management
	4	Explain the concept pages and frames using hardware diagram.	5	L2-CO1-PO1	Paging
	5	Consider a logical address space of 256 pages with a 4-KB page size, mapped onto a physical memory of 64 frames. a. How many bits are required in the logical address? b. How many bits are required in the physical address?	5	L2-CO1-PO1	TLB
	6	Explain demand paging systems.	5	L2-CO1-PO1	Paging

	7	Explain why page fault occurs in a system.	5	L2-CO1-PO1	Paging
	8	Explain the concept of shared pages	5	L2-CO1-PO1	Paging
	9	Explain **Belady's anomaly** in page replacement	5	L2-CO1-PO1	Paging
	10	Explain paging hardware with TLB.	5	L2-CO1-PO1	Paging
6(b)	1	Given memory partitions of 100KB, 500KB, 200KB, 300KB and 600KB (in order), how would each of the First fit, Best-fit and worst -fit algorithms place processes of 212KB, 417KB, 112KB and 426KB (in order)?	15	L4-CO3-PO2	Contiguous Memory Allocation
	2	Given memory partitions of 300KB, 600KB, 350KB, 200KB, 750KB and 125KB (in order), how would each of the First fit, Best-fit and worst -fit algorithms place processes of 115KB, 500KB, 200KB and 375KB (in order)? Which one is more efficient.	15	L4-CO3-PO2	Contiguous Memory Allocation
	3	Analyze how the choice of page size impacts the following aspects of memory management: • (a) Internal fragmentation • (b) Size of page table (c) TLB efficiency Support your answer with numerical examples and diagrams to justify the trade-offs involved in selecting an optimal page size.	15	L4-CO3-PO2	Contiguous Memory Allocation
	4	Compare and analyze the memory management approaches of segmentation and paging: • (a) Explain how each divides memory portions. • (b) Identify the strengths and limitations of each scheme with respect to external/internal fragmentation. (c) Justify why many modern operating systems adopt a segmented-paging scheme, with suitable examples.	15	L4-CO3-PO2	Segmentation/ paging
	5	A computer has a logical address space of 64 KB and a physical memory of 32 KB. The page size is 1 KB. • (a) Analyze how the logical address is divided into page number and page offset. • (b) Determine the number of entries in the page table. (c) Discuss how internal fragmentation and page table size affect overall memory utilization in this system.	15	L4-CO3-PO2	TLB

	6	Analyze how the choice of page size impacts the following aspects of memory management: • (a) Internal fragmentation • (b) Size of page table (c) TLB efficiency	15	L4-CO3-PO2	TLB
	7	The reference string is: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2. The system has 4 frames. • (a) Analyze the number of page faults using FIFO, Optimal, and LRU page replacement algorithms. (b) Compare the results and justify which algorithm is most suitable in this scenario.	15	L4-CO3-PO2	page replacement algorithms.
	8	Analyze the reference string 1, 2, 3, 4, 2,1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 for memory with 5 frames find the page faults for FIFO, Optimal, and LRU page replacement algorithms. which algorithm is most efficient one.	15	L4-CO3-PO2	page replacement algorithms.
	9	A machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB, what is the approximate size of the page table?	15	L4-CO3-PO2	Structure of the Page Table
	10	Compare and contrast inverted page table working to a conventional page table. Explain the role of hashing in searching page table entries.	15	L4-CO3-PO2	Structure of the Page Table
Module 5					
7(a)	1	Apply the system model of resource allocation to represent a scenario where three procsesses compete for two resources each having two instances. Draw the resource allocation graph	10	L3-CO2-PO1	System Model
	2	Apply the deadlock prevention technique to a system with two processes given in the below figure and explain how circular wait can be avoided.  	10	L3-CO2-PO1	Deadlock Prevention
	3	Discuss the necessary conditions for deadlock to occur in a system and illustrate handling of deadlock by using deadlock prevention method.	10	L3-CO2-PO1	Deadlock Characterization
	4	Illustrate recovery from deadlock using process termination and resource preemption.	10	L3-CO2-PO1	Deadlock recovery
	5	Illustrate with an example the usage of Banker's algorithm for deadlock avoidance	10	L3-CO2-PO1	deadlock avoidance
	6	Consider the resource allocation example depicted in the below figure.	10	L3-CO2-PO1	Deadlock avoidance

		a) Show that the four necessary conditions for deadlock hold in this example. b) State a simple rule for avoiding deadlocks in this system.  			
	7	Consider the traffic deadlock depicted in below figure. a) Show the necessary condition for deadlock hold in this example. b) State a simple rule for avoiding deadlocks in this system. 	10	L3-CO2-PO1	Deadlock Avoidance
	8	Given 3 processes A, B and C, three resources x,y and z and following events, i). A requests x ii) A requests y iii) B requests y iv) B requests z v) C requests z vi) C requests x vii) C requests y. Assume that requested resources should always be allocated to the request process if it is available. Draw the resource allocation graph and wait-for graph for the sequence. And mention whether it is a deadlock? If it is, how to recover the deadlock.	10	L3-CO2-PO1	Resource allocation graph
	9	Classify the request and assignment edge in a Resource-allocation graph and construct the graph to show with a deadlock and with a cycle but no deadlock.	10	L3-CO2-PO1	Resource allocation graph
	10	A bank has 10 units of funds and three customers: C1, C2, and C3. Currently, C1 has 3 units, C2 has 2 units, and C3 has 2 units. Their maximum loan needs are 7 units, 5 units, and 3 units respectively. There are 3 units still available. Using Banker's Algorithm, check whether the system is in a safe state or an unsafe state.	10	L3-CO2-PO1	Banker's algorithm

<table>
<tr><td rowspan="7">7(b)</td><td>1</td><td>Analyze the Billing application in a hospital system, that needs access to patient records, while the Diagnosis application needs both patient records and medical imaging devices. These requests may overlap.
Listout the advantages and disadvantages of deadlock prevention and deadlock avoidance in this situation.</td><td>10</td><td>L4-CO3-PO2</td><td>deadlock prevention Vs deadlock avoidance</td></tr>
<tr><td>2</td><td>Analyze the given real-world scenario “3 printer and 4 scanner usage in a lab” to show how deadlocks may occur and suggest a way to handle it.</td><td>10</td><td>L4-CO3-PO2</td><td>Deadlock Characterization + Handling</td></tr>
<tr><td>3</td><td>Consider a system that contains five processes P1,P2,P3,P4 and P5 and the three resource types A, B and C. Following are the resources types: A has 10, B has 5 and the resource types C has 7 instances. Verify whether the system is in safe state or not.
Can a request for (1,0,2) by P1 granted.
<table>
<tr><th>Process</th><th colspan="3">Allocation
A B C</th><th colspan="3">Max
A B C</th><th colspan="3">Available
A B C</th></tr>
<tr><td>P1</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr>
<tr><td>P2</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td colspan="3"></td></tr>
<tr><td>P3</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td colspan="3"></td></tr>
<tr><td>P4</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td colspan="3"></td></tr>
<tr><td>P5</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td colspan="3"></td></tr>
</table></td><td>10</td><td>L4-CO3-PO2</td><td>Banker’ s algo</td></tr>
<tr><td>4</td><td>A system with two printers, one scanner, and three processes is running. Construct a scenario where deadlock occurs and analyze possible recovery strategies. Which is more efficient?</td><td>10</td><td>L4-CO3-PO2</td><td>Deadlock recovery</td></tr>
<tr><td>5</td><td>Construct a Resource Allocation Graph (RAG) for 3 processes and 3 resources. Show a situation that leads to a cycle. Analyze whether this cycle always results in deadlock. Justify your answer</td><td>10</td><td>L4-CO3-PO2</td><td>Resource allocation graph</td></tr>
<tr><td>6</td><td>Consider the following snapshot of a system.
<table>
<tr><th>Process</th><th colspan="4">Allocation
A B C D</th><th colspan="4">Max
A B C D</th><th colspan="4">Available
A B C D</th></tr>
<tr><td>P1</td><td colspan="4">2 0 0 1</td><td colspan="4">4 2 1 2</td><td colspan="4">3 3 2 2</td></tr>
<tr><td>P2</td><td colspan="4">3 1 2 1</td><td colspan="4">5 2 5 2</td><td colspan="4"></td></tr>
<tr><td>P3</td><td colspan="4">2 1 0 3</td><td colspan="4">2 3 1 6</td><td colspan="4"></td></tr>
<tr><td>P4</td><td colspan="4">1 3 1 2</td><td colspan="4">3 6 6 3</td><td colspan="4"></td></tr>
</table>
Answer the following questions using Banker’s Algorithm.
a) Analyze whether the system in a safe state or not.
b) If a request from process P4 (0,0,2,0) arise can it be granted.</td><td>10</td><td>L4-CO3-PO2</td><td>Banker’ s algorithm</td></tr>
<tr><td>7</td><td>In a real computer system, neither the resource available nor the demands of processes for resources are consistent over long periods of time. Resources break or replaced, new processes come and go, and new resources are brought and added to the system. If deadlock is controlled by the Banker’s</td><td>10</td><td>L4-CO3-PO2</td><td>Banker’ s algorithm</td></tr>
</table>

		algorithm, Analyze which of the following changes can be made safely (without introducing possible deadlock), and under what circumstances? a) Increase Available resource (new resource added) b) Decrease available resource (resource permanently removed from system) c) Increase Max for one process (the process needs or wants more resources than allowed). d) Decrease Max for one process (the process decides it does not need that many resources). e) Increase the number of processes. Decrease the number of processes			
	8	Consider the following snapshot of a system. Process | Allocation | Max A B C D | A B C D P0 3 0 1 4 | 5 1 1 7 P1 2 2 1 0 | 3 2 1 1 P2 3 1 2 1 | 3 3 2 1 P3 0 5 1 0 | 4 6 1 2 P4 4 2 1 2 | 6 3 2 5 Using the Banker's algorithm, determine whether or not each of the following states is unsafe. If the state is safe, illustrate the order in which the processes may complete. Otherwise, illustrate why the state is unsafe. a) State 1: Available = (0,3,0,1) b) State 2: Available = (1,0,0,2)	10	L4-CO3-PO2	Banker's algorithm
	9	For the following resource-allocation state, prove that system is not in deadlock state and examine if the request of P2 makes for one additional request for instsance of resource type C ,can be granted or not? Justify your answer. Process | Allocation | Request | Available A B C | A B C | A B C P0 0 1 0 | 0 0 0 | 0 0 0 P1 2 0 0 | 2 0 2 P2 3 0 3 | 0 0 0 P3 2 1 1 | 1 0 0 P4 0 0 2 | 0 0 2	10	L4-CO3-PO2	Deadlock avoidance
	10	Analyse the below snapshot to identify safety sequences of process execution. If P1 request (1,0,1) & in this state P4 request (3,3,0) then identify the safety of the system.	10	L4-CO3-PO2	Banker's algorithm

Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			